Advisor meeting update — October 16, 2026

Progress since October 2 and the manuscript direction

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Main message

We have completed the full-range Single gene-count sensitivity study and requested CpG variability inventory. Prediction stays broadly similar as genes increase; programs are particularly stable from about 5k to 10k genes. Larger CpG fits exposed a numerical failure that is repaired experimentally, but reliable convergence remains the main methodological obstacle.

The benchmark is retained: Single external C=0.627 across five cohorts with $\alpha_F = 0$ (survival does not guide its loading updates). Joint ICGC primary C=0.685 matches expression alone; no incremental predictive benefit is shown.

What increasing the feature count shows

Same 273 TCGA+CPTAC training patients, $K = 7$, no pruning and cold starts; all 24 full-training/CV gene fits converge.

Shared genes	Conditional CV C ± SE	Retained programs
2,064	0.608 ± 0.046	4
5,001	0.577 ± 0.027	4
10,000	0.609 ± 0.040	4
17,900 (all)	0.607 ± 0.019	3

There is no clear predictive gain. Screening/scaling preceded folds; this is conditional CV, not fully nested preprocessing evaluation. No count is adopted.

Expansion	Matched programs	Common-gene loading $\lvert r\rvert$	Risk r
2,064 → 5,001	4	0.115–0.962	0.655
5,001 → 10,000	4	**0.960–0.994**	**0.972**
10,000 → 17,900	3	0.737–0.915	0.667

Strong agreement over 5k–10k does not mean every program is stable across the entire range. Constant/unmatched factors have undefined correlations.

The CpG inventory covers 305,492 raw probes; 305,352 are model-eligible. Observed mean/SD histograms and both requested filter-overlap summaries are complete; no filter or SD cutoff is adopted. The 10k methylation-only fit caps at 500 with zero credible coefficients. Original 25k fitting fails from Cox risk-set underflow. Stable log-space arithmetic removes that error, but the same case still caps at 500 with extreme predictors: training C=0.984 is not a validated improvement. Larger counts and matched CV/joint curves remain unfinished.

What this means for the method and manuscript

Survival-guided loading updates are implemented experimentally, but a fully validated supervised replacement, rank selection and added multimodal value remain unresolved. The next method work addresses prior/objective consistency; software tests alone do not establish a successful scientific remedy.

The full write-up describes the method, benchmark results and provisional biology. Existing evidence supports expression-defined programs and methylation-dominant structure, not its biological cause or added prognostic value. Final labels need annotation/enrichment and composition/batch checks.

Advisor discussion: agree on the manuscript's central contribution and prioritize stable supervision before the remaining matched feature curves. A computational-methods paper with a convincing PDAC evaluation could fit *Bioinformatics Advances*; a stronger statistical-methods emphasis could fit *Statistics in Biosciences*. These are provisional scope-based options, not a submission decision. The current write-up can support drafting now; scientific validation still determines the claims.

Accompanying documents provide the detailed evidence report, the full method/results/biology write-up, and a journal shortlist with sources and remaining manuscript-readiness requirements.